VAMSI KATTERA

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Professional Summary

AI Engineer with hands-on experience building end-to-end machine and deep learning system for real-world applications. Specialized in designing, training, and deploying scalable AI solutions across computer vision and generative AI domains. Proven ability to develop production-ready ML pipelines, integrate LLM-based systems, and deploy real-time AI applications with modern frameworks. Currently focused on building intelligent, low-latency AI system that solve practical problem using computer vision, RAG architectures, and automation-driven AI workflows.

Education

MSc Advanced Computer Science – University of Strathclyde Jan 2025 – Jan 2026
Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence
BSc Computer Science & Engineering – KHIT Andhra, India 2020 – 2024 CGPA:7.66
Relevant Coursework: Python, Data Structures & Algorithms, AI, ML

Skills

Programming: Python, SQL
Machine Learning: ML Pipeline, Pattern Recognition, Data Preprocessing, Model Evaluation, Feature Engineering, Regularization, Cross-Validation, Hyperparameter Tuning, ELT Pipeline
Deep Learning & AI: Neural Networks, CNNs, RNNs, LSTMs, Transformers, LLM Integration, Prompt Engineering, RAG Systems, Embedding Models
Computer Vision: Object Detection, Object Tracking, Video Analytics, OpenCV
AI Agents & MCPs: AI Agent workflows, Tool Integration, LLM Orchestration, Context Management
Backend, APIs: FastAPI, REST API Design, Scalable Backend System, JSON APIs
MLOps & Deployment: Docker, AWS, Render, API Hosting, GitHub CI/CD
Tools & Libraries: TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, OpenAI, Gemini API, Git, GitHub
Mathematical Foundation: Linear Algebra, Calculus, Probability & Statistics

AI Projects

AI Intelligence Traffic Monitoring System
Tech Stack: Python, YOLOv8, FastAPI, OpenCV, Scikit-Learn, Docker, Render

- Built an intelligent traffic monitoring system to automate vehicle detection and traffic analysis from live cam and video feeds, reducing the need for manual observation.
- Designed a computer vision pipeline that detects, counts, and tracks vehicles while monitoring lane usage and traffic density in real time.
- Developed machine learning components to identify unusual traffic patterns and support data-driven traffic management decisions.
- Engineered and deployed a cloud-hosted API, making traffic analytics accessible through a scalable and production-ready service
- Containerized the application with Docker and deployed it online, gaining hands-on experience in end-to-end AI system development, testing, and deployment.

AI vs Real Image Classifier – Deep Learning Computer Vision System
Tech Stack: TensorFlow, Keras, MobileNetV2, Python, Streamlit, AWS EC2

- Built a computer vision system to solve the problem if distinguishing AI-generated images from real-world images, addressing the growing issue of synthetic media detection.
- Designed an end-to-end deep learning pipeline covering data preprocessing, model training, validation, and performance evaluation using transfer learning with MobileNetV2.
- Improved model generalization through regularization techniques, early stopping, and hyperparameter tuning, achieving over 80% classification accuracy.
- Deployed the trained model as a real-time web application using Streamlit, hosted on AWS EC2 for public accessibility and live inference.

Adaptive Coding Challenges – Generative AI & RAG System
Tech Stack: Python, LangChain, Gemini API, Embeddings, Vector Search, Streamlit, AWS

- Designed a context-aware learning system to generate adaptive coding challenges using Retrieval-Augmented Generation (RAG), solving the problem of static and non-personalized practice context.
- Built an end-to-end semantic search pipeline using embeddings and vector search to retrieve relevant context efficiently for LLM-based response generation.
- Improved output quality and relevance through prompt engineering techniques and iterative tuning of retrieval and generation strategies.
- Structured and optimized data flow from ingestion to retrieval, ensuring efficient indexing and fast semantic matching for real-time responses.
- Deployed the full system as a web application using Streamlit and AWS EC2, enabling interactive, real-time AI-driven challenge generation.

Brother Jackey – Autonomous AI Voice Assistant
Tech Stack: Python, Speech Recognition, pyttsx3, Gemini API

- Built a personal AI voice assistant to interact with my laptop hands-free, solving the need for faster and more natural system control through voice commands.
- Integrated speech-to-text and text-to-speech pipelines with an LLM to enable real-time conversational interaction.
- Designed and implemented voice-driven automation, allowing execution of tasks such as opening applications and system control using only spoken commands.
- Improved response accuracy and contextual understanding using prompt engineering and NLP-based processing with Gemini API.
- Structured the system in a modular way to support easy extension of new voice commands and features over time.

Core Competencies

- Problem Solving & System Thinking
- Analytical & Data-Driven Thinking
- Independent Problem Solving
- Collaboration & Technical Communication

Certifications

Deep Learning Specialization – DeepLearning.AI (Coursera)
Mathematics for Machine Learning and Data Science – DeepLearning.AI (Coursera)

Reference available upon Request